START HERE

Where does the score come from?

A Weighted Sum Model helps school staff choose which concern to review first.

Like a school grade
Different parts count for different amounts. ResBack combines four parts into one priority score.

1 Students submit feedback

The system stores the message and its submission date.

↓

2 AI analyzes the message

Gemma through the Gemini API returns sentiment, confidence and topics. ResBack normalizes the topic names.

↓

3 ResBack groups and calculates

Reports about the same topic are counted. Confidence and report age are averaged for negative reports.

↓

4 ResBack adds weighted scores

40% volume + 25% negative ratio + 20% confidence + 15% recency.

↓

5 Staff review the ranked concerns

The highest score appears first. Staff still read the reports.

↓

Two kinds of numbers: feedback supplies the input data; the programmer sets the weights and the 30-day recency rule.

STEP 1 | THE GIVEN DATA

Start with stored report details

These are made-up teaching records. They are not a download of real student feedback.

Example calculation time: 14 October 2026

Assume reports were submitted at the same clock time as the calculation. A 14 September report is exactly 30 days old.

ID	Topic	AI sentiment	AI confidence	Submitted
W1	Wi-Fi	Negative	0.90	14 Sep 2026
W2	Wi-Fi	Negative	0.80	14 Sep 2026
W3	Wi-Fi	Negative	0.95	14 Sep 2026
W4	Wi-Fi	Negative	0.85	14 Sep 2026
W5	Wi-Fi	Negative	1.00	14 Sep 2026
W6	Wi-Fi	Negative	0.90	14 Sep 2026
W7	Wi-Fi	Negative	0.95	14 Sep 2026
W8	Wi-Fi	Negative	0.85	14 Sep 2026
W9	Wi-Fi	Positive	0.95	13 Oct 2026
W10	Wi-Fi	Positive	0.85	14 Oct 2026
C1	Classroom	Negative	0.90	14 Oct 2026
C2	Classroom	Negative	0.90	14 Oct 2026

Where each column comes from

Topic: stored in sentiment_results.concern_topics after AI analysis and topic normalization.

Sentiment and confidence: stored in sentiment_results.sentiment and sentiment_results.confidence from the AI response.

Submission date: stored in feedbacks.created_at when the report is saved.

Report age: calculated from the current time minus the submission time. The ranking code uses elapsed days.

STEPS 2 AND 3 | COUNT THE REPORTS

Complaint amount and percentage

Complaint amount means negative-feedback volume. It is not the total number of all reports.

Topic	All reports T	Negative N
Wi-Fi / Internet	10	8
Classroom / Room	2	2

A Complaint amount score V

Count negative reports for every topic. The largest negative count is **M = 8**. The code uses at least 1 for M to keep the calculation safe.

```
V = ln(1 + N) / ln(1 + M)
```

Wi-Fi: ln(9) / ln(9) = **1.00**
Classroom: ln(3) / ln(9) = **0.50**

ln means natural logarithm: a calculator operation that softens large count differences. Classroom has one quarter as many complaints, but receives half the volume score.

B Negative percentage score P

```
P = N / T
```

Wi-Fi: 8 / 10 = **0.80** = 80% negative
Classroom: 2 / 2 = **1.00** = 100% negative

Wi-Fi feedback distribution

One bar = 10 Wi-Fi reports. Source: the illustrative records on page 2.

Confidence and recency

Only negative reports are used for these two averages.

C Average negative AI confidence

Take the confidence values attached to W1 through W8. Do not include the two positive Wi-Fi reports.

```
C = sum of negative confidence values / N
```

0.90 + 0.80 + 0.95 + 0.85 + 1.00 + 0.90 + 0.95 + 0.85 = **7.20**
C = 7.20 / 8 = **0.90**

The model returns these values; the ranking code does not invent them. Confidence describes certainty about the sentiment label, not proven accuracy, truth or seriousness.

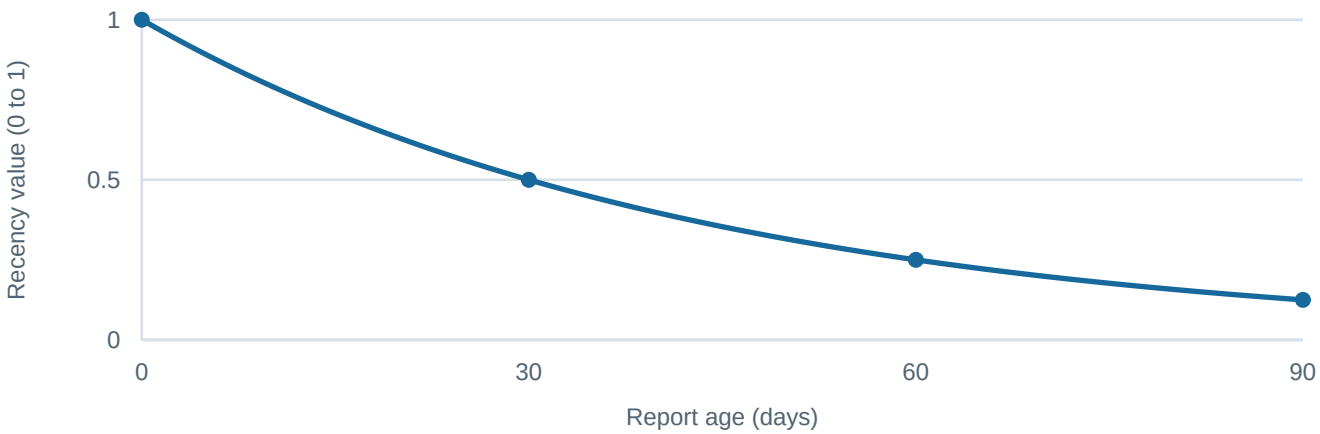
D Average negative recency R

```
Each report: r = 2^(-age in days / 30)
```

For each Wi-Fi complaint: age = 30 days.
r = 2^(-30 / 30) = 2^(-1) = **0.50**
All eight have the same age, so R = (8 x 0.50) / 8 = **0.50**.

If ages differ, calculate each report separately, then average the results. The code writes the same formula as exp(-ln(2) x age / 30).

How one report loses recency value



Source: actual code rule. After 30 days, only the recency value halves; the whole final score does not halve.

STEP 6 | MULTIPLY AND ADD

Turn four scores into 85.5

All four inputs use the same 0-to-1 scale before the weights are applied.

```
Score = 100 x (0.40V + 0.25P + 0.20C + 0.15R)
```

The programmer sets these weights in ConcernRankingService.php. They total 1.00 (100%). They are fixed choices, not weights learned by the AI.

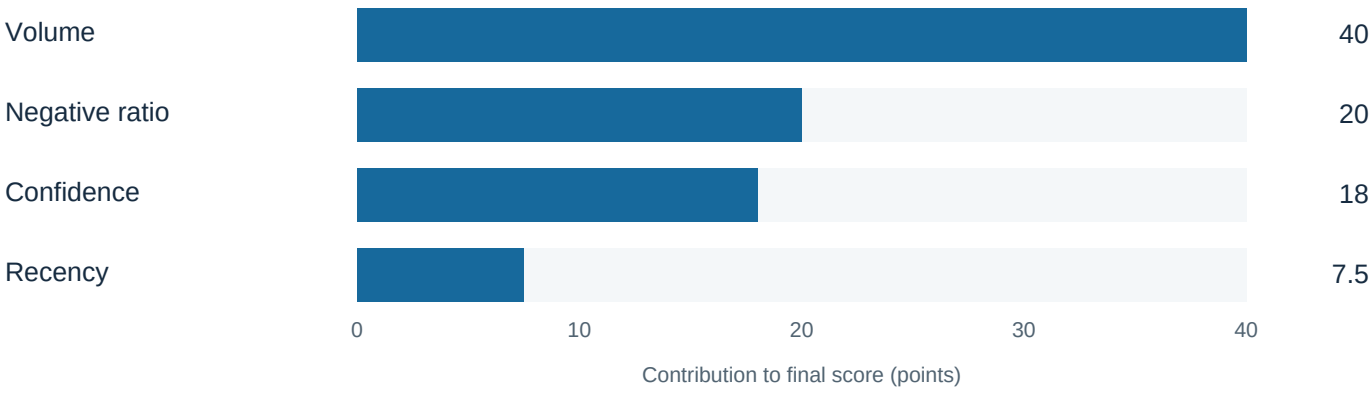
Factor	Wi-Fi value	Weight	Points out of 100
Volume V	1.00	40%	100 x 1.00 x 0.40 = 40
Negative ratio P	0.80	25%	100 x 0.80 x 0.25 = 20
Confidence C	0.90	20%	100 x 0.90 x 0.20 = 18
Recency R	0.50	15%	100 x 0.50 x 0.15 = 7.5

Add the four contributions

40 + 20 + 18 + 7.5 = 85.5 points

The code rounds the final score to one decimal place.

See what each factor contributes



Rows: scoring factors. Source: the Wi-Fi example on pages 2 to 4. Maximum contributions are 40, 25, 20 and 15 points.

A score of 85.5 means priority for review. It does not mean 85.5% accurate or 85.5% dangerous.

STEP 7 | RANK AND REVIEW

Compare the concerns

The algorithm repeats the same calculation for every topic in the selected feedback set.

Calculate Classroom using its two reports

V = 0.50: ln(3) / ln(9).
P = 1.00: both reports are negative.
C = 0.90: (0.90 + 0.90) / 2.
R = 1.00: both reports were submitted today.

```
Classroom = 20 + 25 + 18 + 15 = 78.0
```

Rank	Concern	Final score	Evidence
1	Wi-Fi / Internet	85.5	10 total reports
2	Classroom / Room	78.0	2 total reports: Low evidence

What the ranking means

Wi-Fi comes first in this example. Classroom has a higher negative percentage and newer reports, but Wi-Fi earns more volume points. The weights decide how these differences combine.

Rules that help you explain it correctly

Ties: higher negative count comes first; if still tied, topic names are sorted alphabetically.

Low evidence: fewer than 3 total reports adds a label. It does not lower the score. A topic with no negative reports scores 0.

Filters: category, language and date filters can change the counts, averages and largest count M. The dashboard shows the top 10.

Multiple topics: one feedback entry can count in more than one topic. Its overall sentiment and confidence are reused for those topics.

Human decision: the formula has no separate harm-severity factor. Staff review the actual messages before acting.

Follow the code in your project

app/Services/SentimentService.php: analyze() and parseResult() obtain, validate and store the AI output.
app/Services/ConcernTopicService.php: normalize() groups topic names.
app/Services/ConcernRankingService.php: rank() counts, scores and sorts.
app/Http/Controllers/DashboardController.php: index() selects the top 10.

Remember: given data -> four calculated inputs -> fixed weights -> final score -> ranking -> staff review.